



PHYSICAL AND STRUCTURAL GEOLOGY

GS 21/L

Sedimentary Rocks



Sedimentary Rocks

Most of Earth's solid surface consists of either sediment or sedimentary rock. About 75 percent of land areas are covered by sediments and sedimentary rocks. Sedimentary rocks contain fossils, which are vital tools in the study of the geologic past. This group of rocks provides geologists with much of the basic information they need to reconstruct the details of Earth history.



Sedimentary Rocks

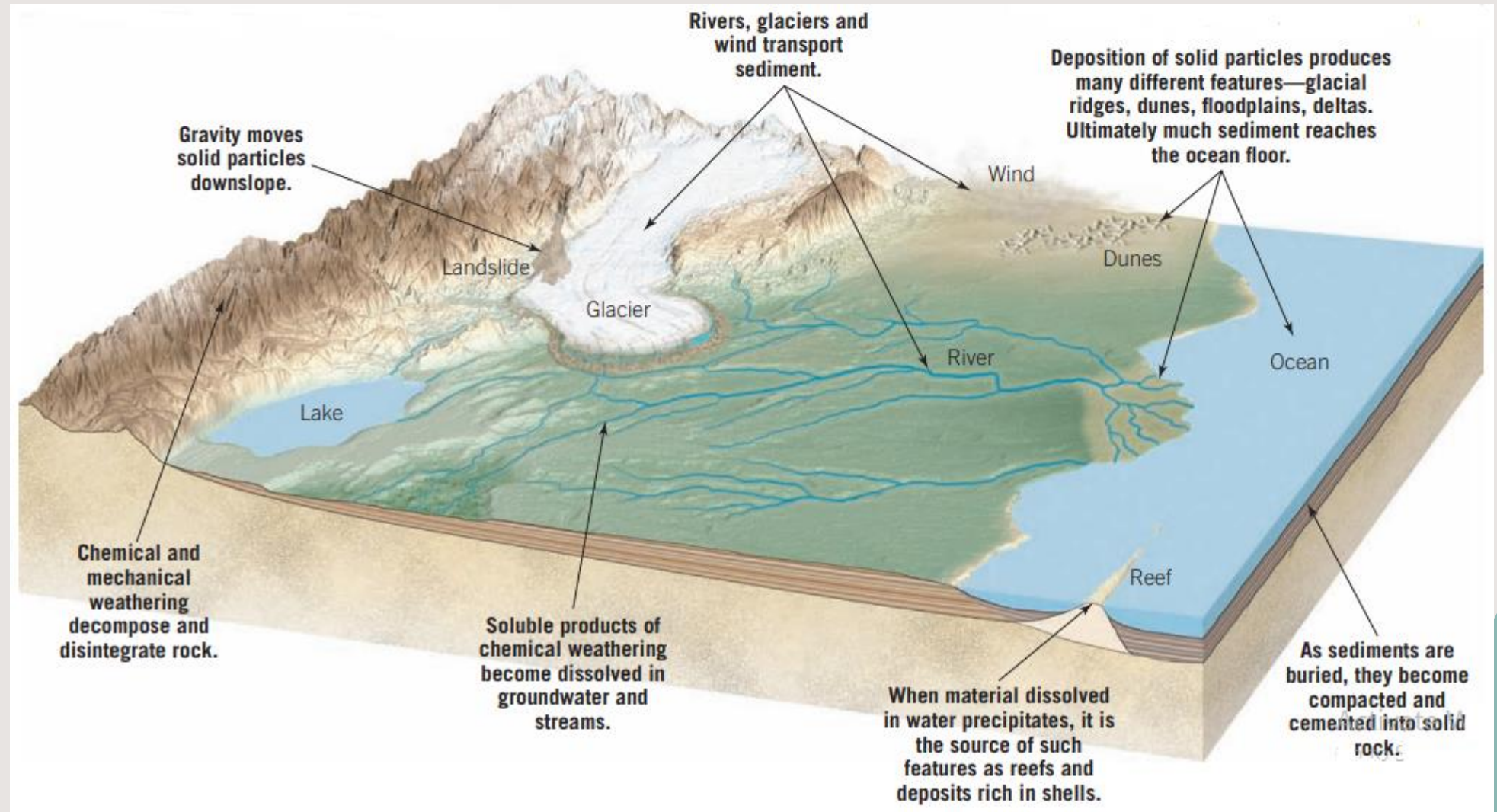


Figure 8.1 The Big picture

An outline of the Portion of the rock cycle that pertains to the formation of sedimentary rocks.

Sedimentary Rocks

Formation of such type of rock begins with:

WEATHERING



EROSION



DEPOSITION



DIAGENESIS/LITHIFICATION

Sedimentary Rocks

Weathering - involves the physical disintegration and chemical decomposition of preexisting igneous, metamorphic, and sedimentary rocks. Weathering generates a variety of products, including various solid particles and ions in solution. These are the raw materials for sedimentary rocks.

Soluble constituents are carried away by runoff and groundwater. Solid particles are frequently moved downslope by gravity, a process termed **mass wasting**, before running water, groundwater, wind, and glacial ice remove them. Transportation moves these materials from the sites where they originated to locations where they accumulate.

Deposition of solid particles occurs when wind and water currents slow down and as glacial ice melts. The word **sedimentary** actually refers to this process. It is derived from the Latin **sedimentum**, which means “to settle,” a reference to solid material settling out of a fluid (water or air).

Sedimentary Rocks

The deposition of material dissolved in water is not related to the strength of water currents. Rather, ions in solution are removed when chemical or temperature changes cause material to crystallize and precipitate or when organisms remove dissolved material to build shells.

As deposition continues, older sediments are buried beneath younger layers and gradually converted to sedimentary rock (lithified) by compaction and cementation. This and other changes are referred to as **diagenesis** (**dia** = change; **genesis** = origin), a collective term for all the changes (short of metamorphism) that take place in texture, composition, and other physical properties after sediments are deposited.

Sedimentary Rocks

Sedimentary Rock Classification:

1. Detrital - an accumulation of material that originates and is transported as solid particles derived from both mechanical and chemical weathering.

2. Chemical - soluble material produced largely by chemical weathering. When these ions in solution are precipitated by either inorganic or biological processes, the material is known as ***chemical sediment***, and the rocks formed from it are called ***chemical sedimentary rocks***.

3. Organic - from the carbon-rich remains of organisms. The primary example is coal. This black combustible rock consists of organic carbon from the remains of plants that died and accumulated on the floor of a swamp.

Sedimentary Rocks

Sedimentary Rock Classification

1. Detrital Sedimentary Rocks

Other common minerals in detrital rocks are Feldspars and micas. Because chemical weathering rapidly transforms these minerals into new substances, their presence in sedimentary rocks indicates that erosion and deposition were fast enough to preserve some of the primary minerals from the source rock before they could be decomposed.

Particle size is not only a convenient method of dividing detrital rocks, the sizes of the component grains also provide useful information about environments of deposition. Currents of water or air sort the particles by size; the stronger the current, the larger the particle size carried.

Figure 8.2 Particle size categories

Particle size is the primary basis for distinguishing among various detrital sedimentary rocks.

(Photos by E. J. Tarbuck)

Size Range (millimeters)	Particle Name	Common Name	Detrital Rock
>256	Boulder	Gravel	 Conglomerate
64-256	Cobble		
4-64	Pebble		
2-4	Granule		
1/16-2	Sand	Sand	 Sandstone
1/256-1/16 <1/256	Silt Clay	Mud	 Shale Mudstone or Siltstone
			

Sedimentary Rocks

Sedimentary Rock Classification

1. Detrital Sedimentary Rocks

A. Shale - a sedimentary rock consisting of silt- and clay-size particles. These fine-grained detrital rocks account for well over half of all sedimentary rocks. The particles in these rocks are so small that they cannot be readily identified without great magnification, and this makes shale more difficult to study and analyze than most other sedimentary rocks.

The tiny grains in shale indicate that deposition occurs as the result of gradual settling from relatively quiet, nonturbulent currents.

Siltstone – without fissility

Shale – exhibits fissility

Mudstone – breaks into chunks

Figure 8.3 Shale—The most abundant sedimentary rock

Dark shale containing plant remains is relatively common.
(Photo by E. J. Tarbuck)



Sedimentary Rocks

Sedimentary Rock Classification

1. Detrital Sedimentary Rocks

B. Sandstone - is the name given rocks in which sand-size grains predominate.

Parameters that provide clues on the formational environment of sandstones:

1. Sorting: well sorted=long provenance, exposed to uniform energy for longer time; poorly sorted= short provenance, possibly exposed to abrupt change in energy upon deposition (high to low)

2. Shape: Rounded/High Sphericity- long provenance, high energy environment; Angular/Low sphericity= short provenance, low energy environment.

3. Mineral composition:

High Qtz content = long provenance;

High Ferromagnesian/Feldspars = short provenance

Figure 8.4 Quartz sandstone

After shale, sandstone is the most abundant sedimentary rock.



Sedimentary Rocks

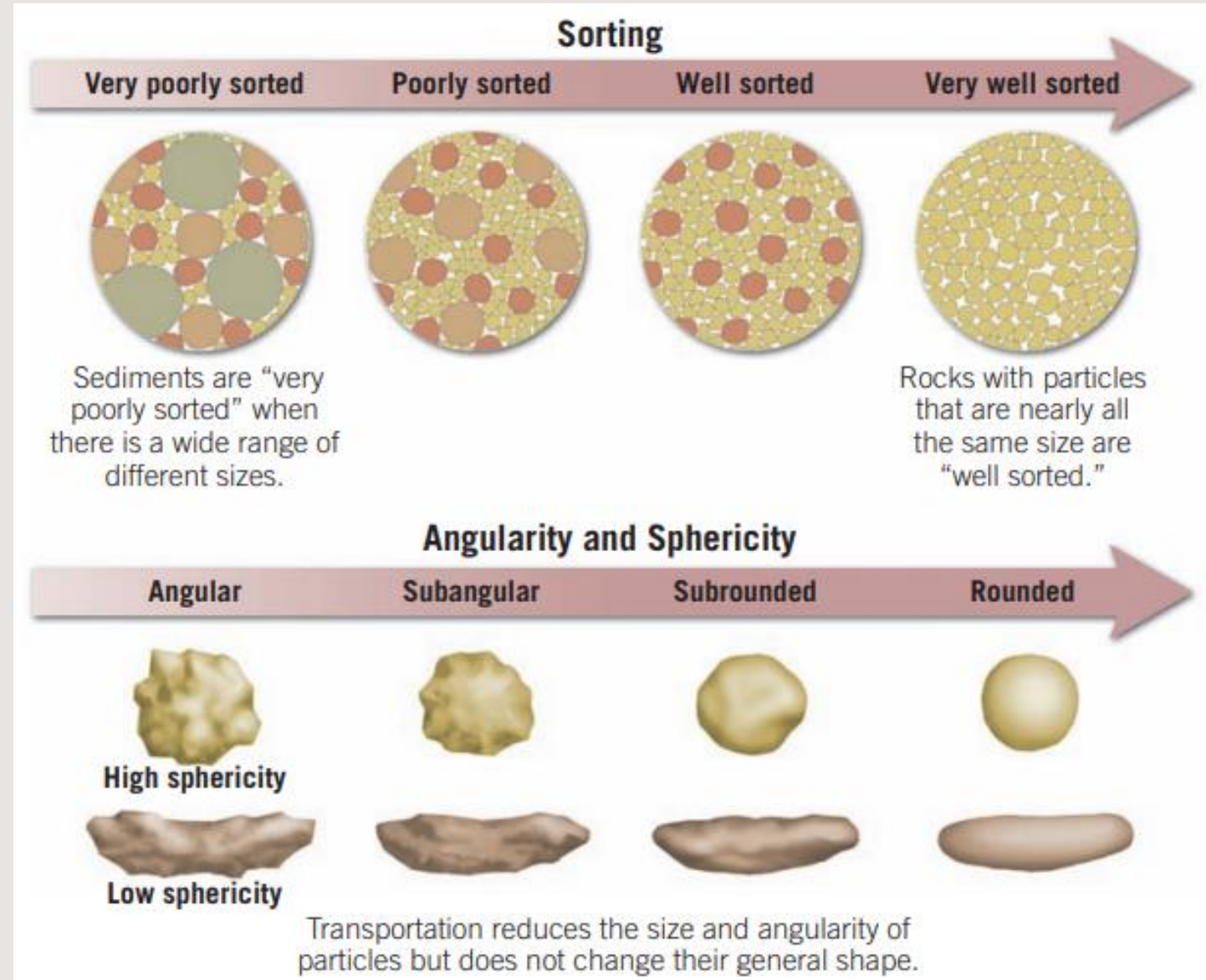
Sedimentary Rock Classification

1. Detrital Sedimentary Rocks

B. Sandstone

Figure 8.5 Sorting and particle shape

Sorting refers to the range of particle sizes present in a rock. Geologists describe a particle's shape in terms of its angularity (degree to which edges and corners are rounded) and sphericity (how close the shape is to a sphere).



Sedimentary Rocks

Sedimentary Rock Classification

1. Detrital Sedimentary Rocks

B. Sandstone

Types of Sandstones:

Quartz Sandstone - the predominant mineral in most sandstones due to its durability.

Arkose - a sandstone which contains appreciable quantities of feldspar (25 percent or more)

Graywacke - this dark-colored rock contains abundant rock fragments and matrix. The poor sorting and angular grains characteristic of graywacke suggest that the particles were transported only a relatively short distance from their source area and then rapidly deposited.

C. Conglomerate and Breccia - consists largely of gravels. These particles can range in size from large boulders to particles as small as peas. Because large particles abrade and become rounded very rapidly during transport, the pebbles and cobbles in a breccia indicate that they did not travel far from their source area before they were deposited.

Sedimentary Rocks

Sedimentary Rock Classification

1. Detrital Sedimentary Rocks

C. Conglomerate and Breccia



Figure 8.6 Conglomerate

The gravel-size particles in this rock are rounded.
(Photo by E. J. Tarbuck)



Figure 8.7 Breccia

The gravel-size sediments in this rock are sharp and angular.
(Photo by E. J. Tarbuck)

Sedimentary Rocks

Sedimentary Rock Classification

2. Chemical Sedimentary Rocks - chemical sediments derive from ions that are carried in solution to lakes and seas. This material does not remain dissolved in the water indefinitely, however. Some of it precipitates to form chemical sediments. These become rocks such as limestone, chert, and rock salt.

This precipitation of material occurs in two ways:

Inorganic (in = not, organicus = life) processes such as evaporation and chemical activity can produce chemical sediments. (i.e. travertine, oolitic limestones)

Organic (life) processes of water-dwelling organisms also form chemical sediments, said to be of biochemical origin. (i.e. reefal limestones, coquina, chalk)

a. Limestone - representing about 10 percent of the total volume of all sedimentary rocks, limestone is the most abundant chemical sedimentary rock. It is composed chiefly of the mineral calcite (CaCO_3) and forms either by inorganic means or as a result of biochemical processes.

b. Dolostone - a rock composed of the calcium-magnesium carbonate mineral **dolomite** [$\text{CaMg}(\text{CO}_3)_2$]. It appears that significant quantities of dolostone are produced when magnesium-rich waters circulate through limestone and convert calcite to dolomite by the replacement of some calcium ions with magnesium ions (a process called dolomitization).

Delicate calcite crystals forming in a drop of water at the tip of soda straw stalactite. The formation of crystals is triggered when some carbon dioxide escapes from the water drop.

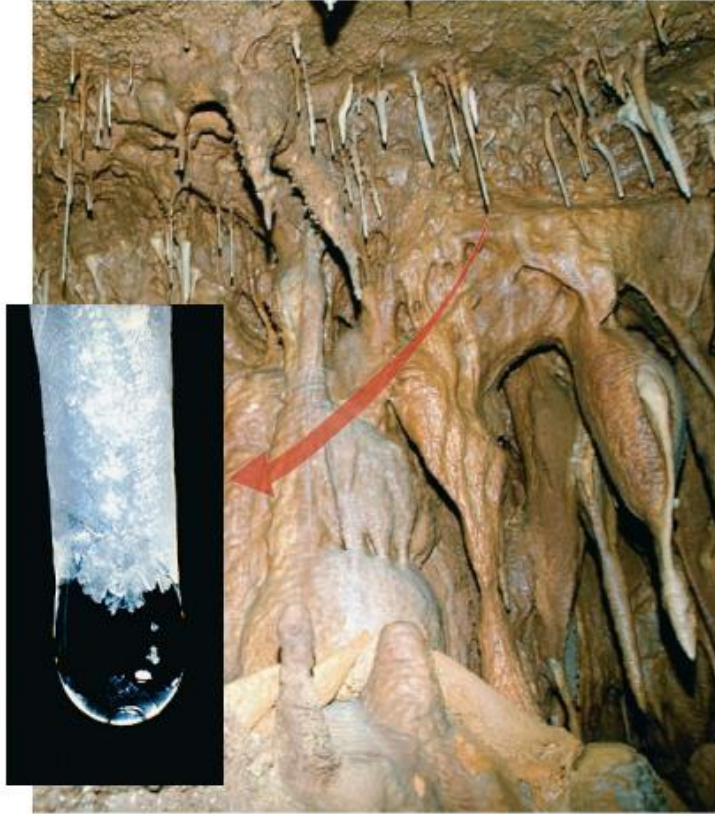


Figure 8.8 Cave deposits

An example of a chemical sedimentary rock with an inorganic origin. (Photo by Guillen Photography/Alamy Images; inset photo by Dante Fenolio/Photo Researchers, Inc.)



Figure 8.9 Coquina

This variety of limestone consists of shell fragments; therefore, it has a biochemical origin. (Rock sample photo by E. J. Tarbuck; beach photo by Donald R. Frazier Photolibrary, Inc./Alamy Images)

Sedimentary Rocks

Sedimentary Rock Classification

2. Chemical Sedimentary Rocks

c. Chert - is a name used for a number of very compact and hard rocks made of microcrystalline quartz (SiO_2). Colors of such varies depending on the impurities it contain.

Types of Chert:

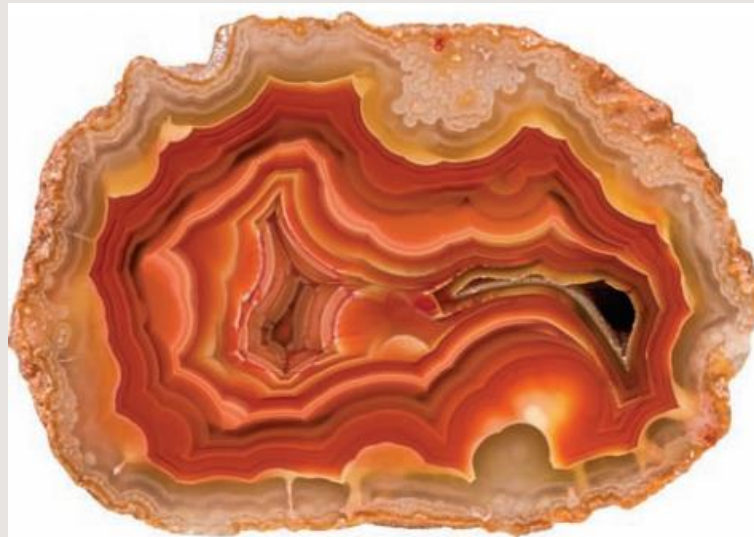
1. **Flint** - a dark colored microcrystalline quartz results from the organic matter it contains.
2. **Jasper** - a red colored microcrystalline quartz, gets its bright color from iron oxide.
3. **Agate** - banded microcrystalline quartz.
4. **Petrified wood** - is made when silica-rich material such as volcanic ash buries trees. Ancient wood replaced with microcrystalline quartz.



Flint



Jasper



Agate



Petrified wood

Sedimentary Rocks

Sedimentary Rock Classification

2. Chemical Sedimentary Rocks

d. Evaporites - deposits formed by precipitation of water from a basins.

Minerals commonly precipitated as evaporites:

Halite (sodium chloride, NaCl), the chief component of rock salt and gypsum (hydrous calcium sulfate, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), the main ingredient of rock gypsum.

3. Organic Sedimentary Rock

Coal – it is a rock that burns.

- usually forms in stagnant swamps where conditions are oxygen deficient.
- composed of organic matter like plant fragments that are chemically altered but still identifiable.

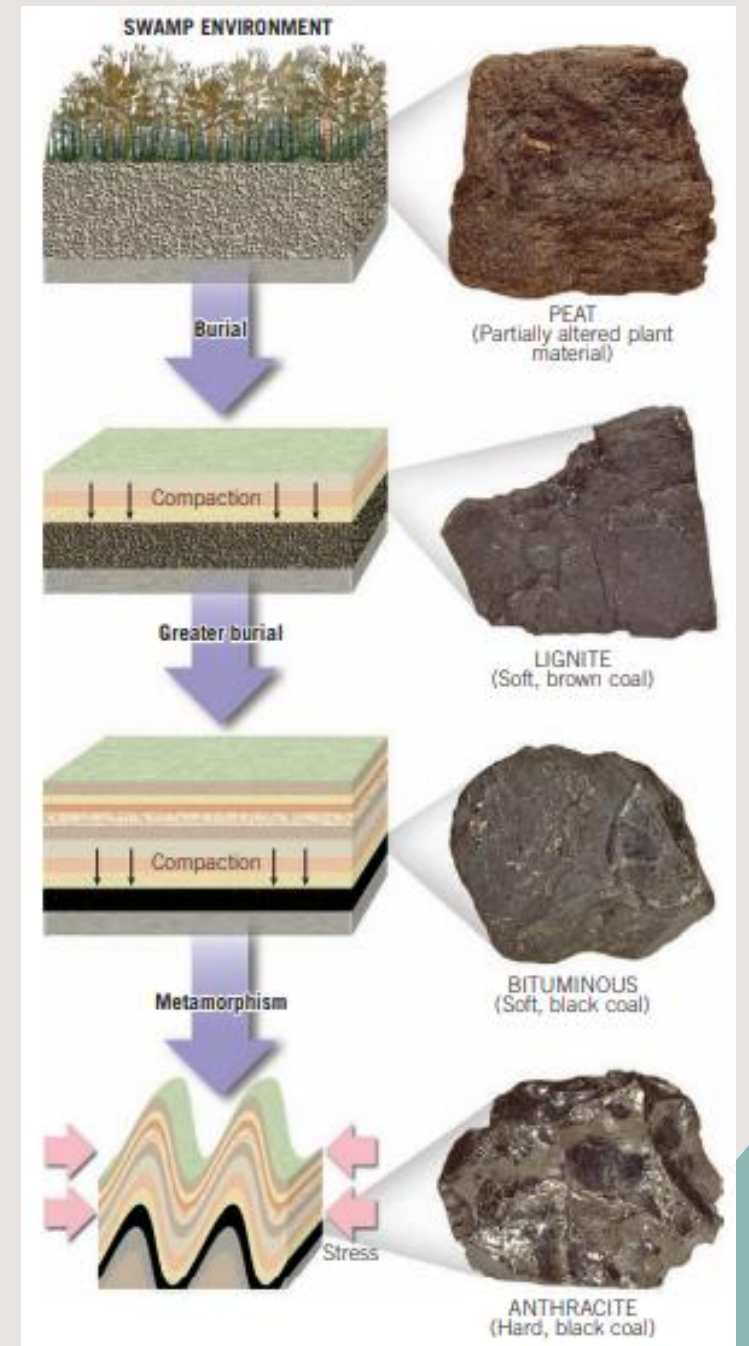
Sedimentary Rocks

Sedimentary Rock Classification

3. Organic Sedimentary Rock

Figure 8.10 From plants to coal

Successive stages in the formation of coal. (Photos by E. J. Tarbuck)



Sedimentary Rocks

Sedimentary Rock Classification

3. Organic Sedimentary Rock

Stages in the formation of coal:

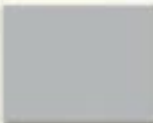
Accumulation of plant remains – initial stage. Special conditions are required for such accumulations because dead plants readily decompose when exposed to the atmosphere or other oxygen-rich environments. One important environment that allows for the buildup of plant material is a swamp.

Formation of peat - partial decomposition of plant remains in an oxygen-poor swamp creates a layer of peat, a soft brown material in which plant structures are still easily recognized. With shallow burial, peat slowly changes to lignite, a soft brown coal.

Formation of lignite and bituminous coal - the higher temperatures bring about chemical reactions within the plant materials and yield water and organic gases (volatiles). As the load increases from more sediment on top of the developing coal, the water and volatiles are pressed out, and the proportion of fixed carbon (the remaining solid combustible material) increases. The greater the carbon content, the greater the coal's energy ranking as a fuel.

Formation of anthracite coal - when sedimentary layers are subjected to the folding and deformation associated with mountain building, the heat and pressure cause a further loss of volatiles and water, thus increasing the concentration of fixed carbon. This metamorphoses bituminous coal into anthracite, a very hard, shiny, black metamorphic rock.

Sedimentary Rocks

Detrital Sedimentary Rocks			
Clastic Texture (particle size)		Sediment Name	Rock Name
Coarse (over 2 mm)		Gravel (Rounded particles)	Conglomerate
		Gravel (Angular particles)	Breccia
Medium (1/16 to 2 mm)		Sand	Sandstone (Arkose)*
Fine (1/16 to 1/256 mm)		Mud	Siltstone
Very fine (less than 1/256 mm)		Mud	Shale or Mudstone

*If abundant feldspar is present the rock is called Arkose.

Figure 8.11 Identification of sedimentary rocks
The main criterion for naming detrital rocks is particle size. The primary basis for naming chemical and organic sedimentary rocks is their composition.

Chemical and Organic Sedimentary Rocks			
Composition	Texture	Rock Name	
Calcite, CaCO ₃	Nonclastic: Fine to coarse crystalline	Crystalline Limestone	Bioc hem ical
		Travertine	
	Clastic: Visible shells and shell fragments loosely cemented	Coquina	
	Clastic: Various size shells and shell fragments cemented with calcite cement	Fossiliferous Limestone	
	Clastic: Microscopic shells and clay	Chalk	ne
Quartz, SiO ₂	Nonclastic: Very fine crystalline	Chert (light colored) Flint (dark colored) Jasper (red) Agate (banded)	
Gypsum CaSO ₄ •2H ₂ O	Nonclastic: Fine to coarse crystalline	Rock Gypsum	
Halite, NaCl	Nonclastic: Fine to coarse crystalline	Rock Salt	
Altered plant fragments	Nonclastic: Fine-grained organic matter	Bituminous Coal	

Sedimentary Rocks

Sedimentary Rock Structures

Beds/Strata – sedimentary rocks form as layer upon layer of sediment accumulates in various depositional environments.

Laminations – layering thinner than 10 mm. Layering within a sediment due to fluctuations in flow conditions. The layering may be expressed by variations in particle size, sediment composition or in grain alignment.

Cross Beddings - layers of sedimentary rock that are inclined to the horizontal.

Ripple Marks - are small waves of sand that develop on the surface of a sediment layer by the action of moving water or air.

Mud Cracks - indicate that the sediment in which they were formed was alternately wet and dry.

Fossils - the remains or traces of prehistoric life, are important inclusions in sediment and sedimentary rocks.

Sedimentary Rocks

Resources From Sedimentary Rocks

Salt – Halite (NaCl), Non clastic: Fine to coarse crystalline

Gypsum – $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, Non clastic: Fine to coarse crystalline

Gravel – Conglomerate, Breccia, Coarse (over 2 mm)

Oil/gas – Altered plant fragments, Non clastic: Fine-grained organic matter

Cement (limestone) – Calcite (CaCO_3)

Reference

- Tarbuck, Lutgens and Tasa (2014), Earth: An Introduction to Physical Geology 11th ed., Pearson Education, Inc